

RouterMAS: Value-Preserving Watermarking for Multi-Agent Routing

crcr0.github.io/RouterMAS

project page · interactive story
case studies · extended abstract

A secret key moves routing probability only inside the freedom left after continuation utility and cost are fixed; the same marked law yields an anytime-valid verifier.

Google · xAI · Cornell University · Columbia University

1 The router is the asset

Two systems built from the same models can embody different orchestration policies, so the routing policy is what an owner wants to attribute. But a delegation is a functional decision: it changes the history on which every later choice, retry and cost depends.

We require the marked router to preserve expected task utility and agent-call cost **after every admissible execution prefix**, so gains on easy prefixes cannot pay for losses on hard ones.

Red/green delegation bias ✗

Blind to what each delegation enables later; continuation values drift.

Behavioral marks ✗

Aggregate utility kept; prefix-wise continuation values are not in the contract.

Forced secret paths ✗

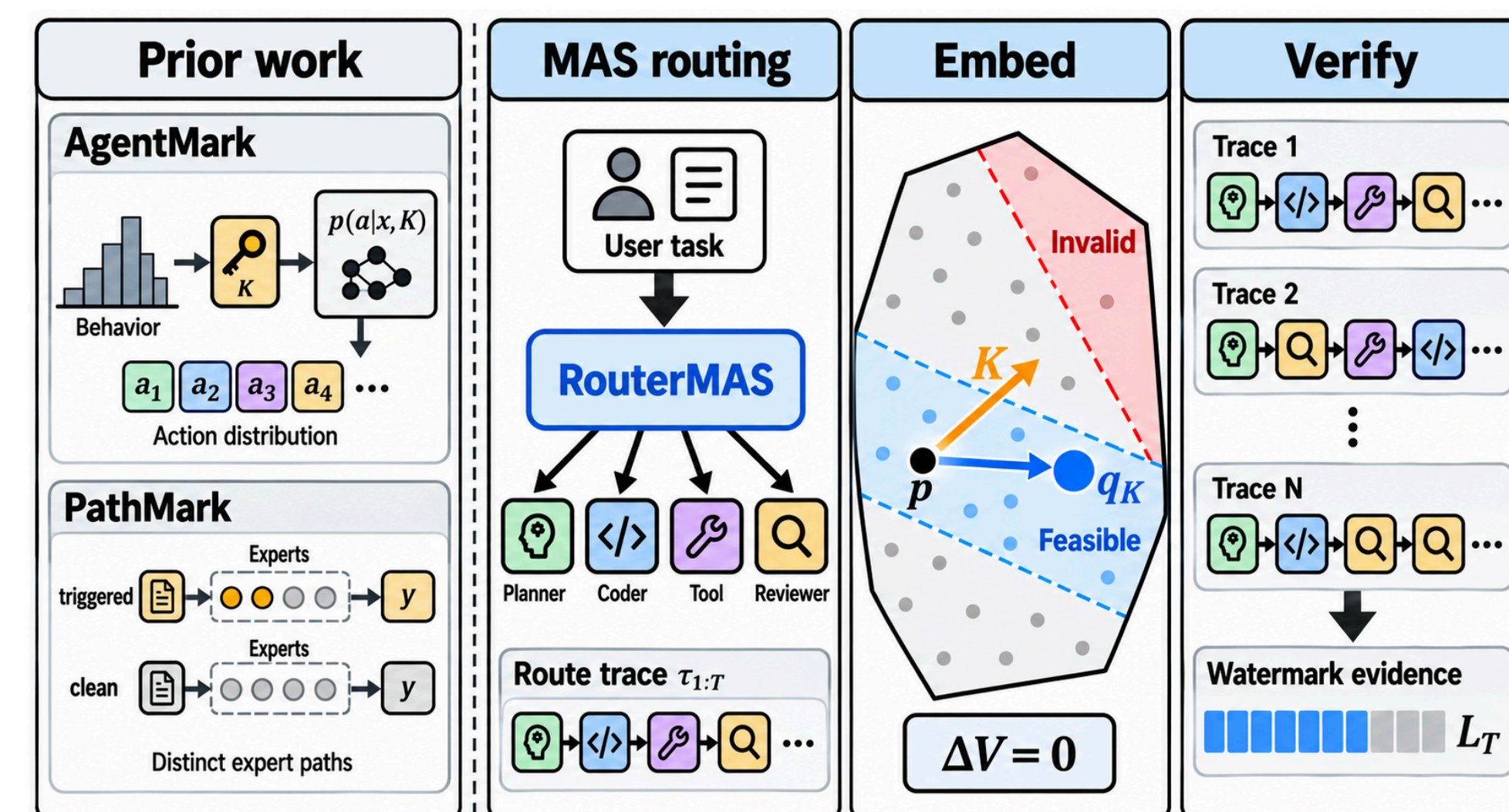
Detectable because they send work where the original router would not.

RouterMAS

Derive the admissible carrier from continuation values first; abstain where none exists.

Core question. Which routing variations preserve the continuation values a deployment must keep, and how can they carry a verifiable signature with controlled intervention and estimation cost?

Overview



Behavioral marks act on agent actions and path marks on expert routes; RouterMAS moves the system-level routing law p to q_K inside the continuation-value-preserving region ($\Delta V = 0$), and delegation traces accumulate likelihood-ratio evidence L_T .

Setting

Frozen agents, tools and executor. At history h : legal delegations $\mathcal{A}_h$, reference law $p_h > 0$, and for each protected return j the continuation value $Q_j^p(h, a)$ — the expected return after delegating to a and then following p to termination. Contract: $V_j^q(h) = V_j^p(h)$ at every admissible prefix.

2 Method

1 Continuation conservation defines the carrier

Audit each legal delegation by complete counterfactual continuations under the unchanged reference router.

$$V_j^q(h) = V_j^p(h) \forall h \iff (q_h - p_h)^\top Q_j^p(h) = 0 \forall h$$

Proposition 1 (prefix-wise conservation)

$$\dim \mathcal{F}_h = |\mathcal{A}_h| - \text{rank } B_h, \quad B_h = [1^\top; Q_1^p(h)^\top; \dots; Q_m^p(h)^\top]$$

feasible family; full rank or a row-space key $\Rightarrow$ return p

2 Least-change keyed encoding

Information projection inside $\mathcal{F}_h$: the smallest relative-entropy change for its key-score gain, an exponential tilt.

$$q_K(a) \propto p(a) \exp((s_K(a) - \lambda^\top \widehat{Q}_a)/\eta)$$

multipliers λ enforce conservation; safe scaling $q_K = p + \rho \delta$ charges estimation error

3 Sequential verification

Rebuild p and q_K from frozen snapshots and sum log-likelihood ratios over observed delegations.

$$L_T = \sum_{t < T} \log \frac{q_{K, h_t}(a_t)}{p_{h_t}(a_t)}, \quad P_p(\sup_t L_t \geq \log \frac{1}{\alpha}) \leq \alpha$$

Ville's inequality: anytime-valid, stop at the first crossing

3 One geometry for embedding and evidence

$$s_K^\top g = g^\top D^{-1} g = \|PD^{1/2} s_K\|^2, \quad g = D^{1/2} PD^{1/2} s_K$$

$D = \text{diag}(p)$, $P = \text{projector onto } \ker(B D^{1/2})$

$$\text{gain} = \eta^{-1} s_K^\top g + O(\eta^{-2}) \cdot \text{evidence} = \frac{1}{2} \eta^{-2} s_K^\top g + O(\eta^{-3})$$

Corollary 1 — zero projection makes both inert

The projection that selects the least-change keyed update also fixes its leading-order verification information. Halving the final update halves the gain and quarters the evidence.

Error budget

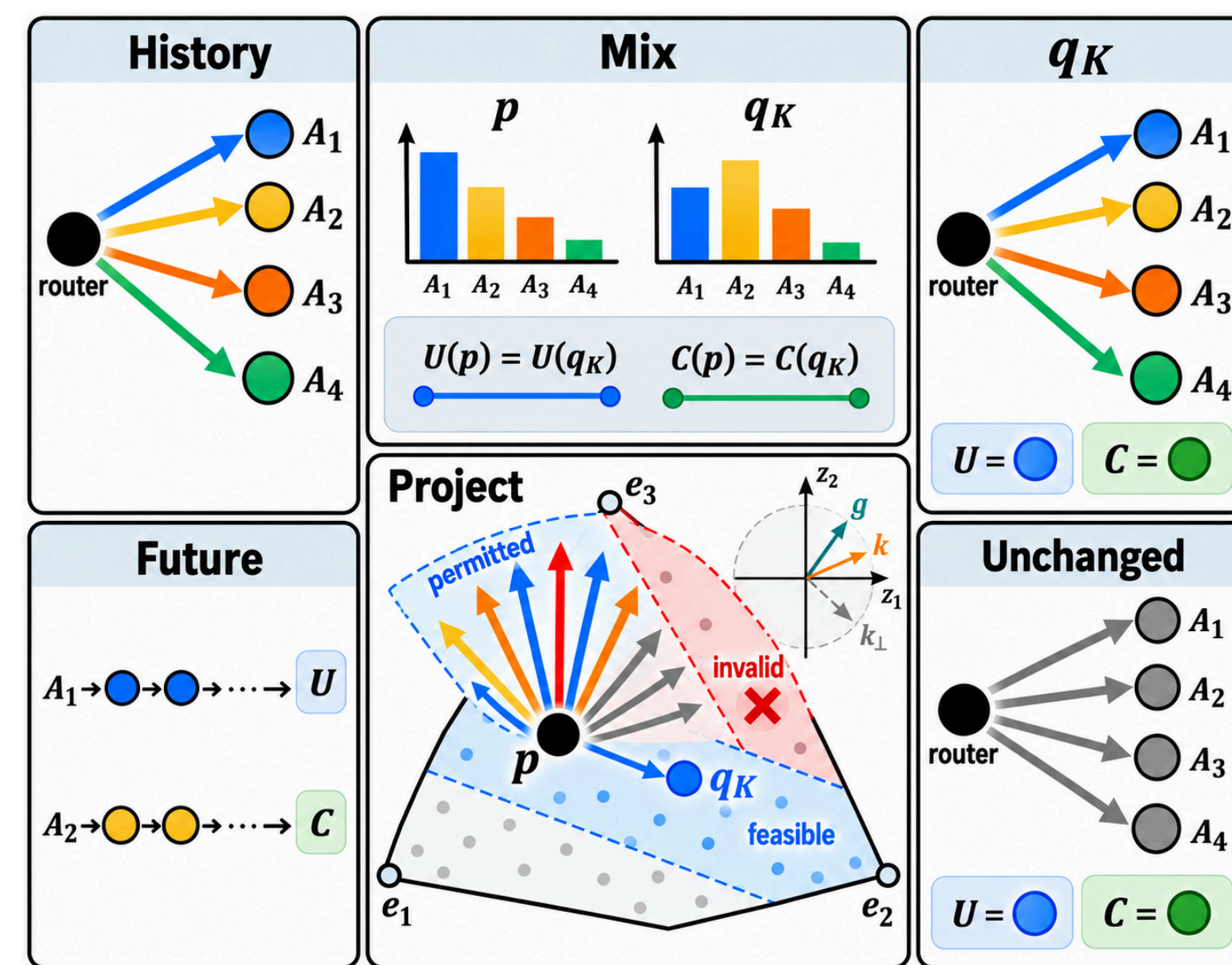
$$|J_j(\tilde{q}) - J_j(p)| \leq \mathbb{E}_{\tilde{q}} \sum_{t < H} [b_{j, h_t} + \epsilon_{j, h_t} \|\tilde{q}_{h_t} - p_{h_t}\|_1]$$

Audit uncertainty, solver residuals and controller compilation enter one return budget along the implemented policy's own visits.

Algorithm 1 · embedding

1. $(\mathcal{A}, p) \leftarrow \text{LegalRouting}(h)$; $(\widehat{Q}, \epsilon) \leftarrow \text{ContinuationAudit}(h, M)$
2. $B \leftarrow [1^\top; \widehat{Q}^\top]$, $D \leftarrow \text{diag}(p)$, $P \leftarrow \text{Proj ker}(BD^{1/2})$
3. $g \leftarrow D^{1/2} PD^{1/2} s_K$; if $g = 0$ or error unbounded **return** p
4. $q^* \leftarrow \arg \max_{q \in \mathcal{F}} s_K^\top (q - p) - \eta \text{KL}(q \| p)$
5. $\rho \leftarrow \min\{1, \min_j \tau_j / (b_j + \epsilon_j \|\delta\|_1)\}$; **return** $p + \rho(q^* - p)$

One history, two values held fixed



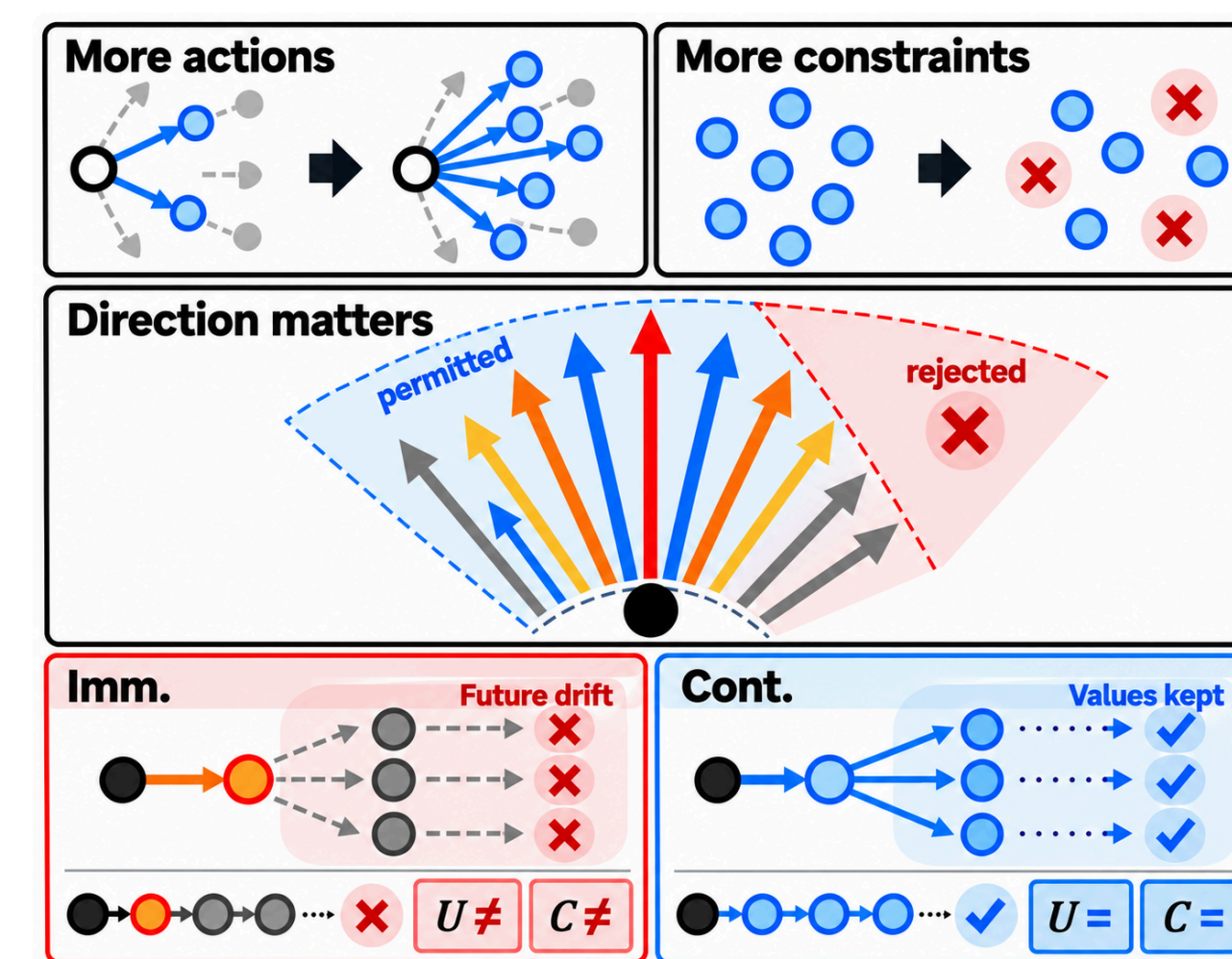
p and q_K mix the delegations differently with equal continuation utility and cost; value-changing directions are excluded, and histories without a usable direction keep the reference router.

Sequential verification (%)

T	FPR $\alpha=.01$	TPR $\alpha=.01$	FPR $\alpha=.001$	TPR $\alpha=.001$
32	0.36	50.81	0.02	18.88
64	0.67	88.24	0.03	69.73
128	0.75	99.30	0.04	97.46
256	0.75	100.00	0.05	99.99

10,000 independent trace groups per law; running-maximum rejection. Reference false-positive rates stay below α at every budget.

Why continuation, not immediate



More legal delegations expose freedom, more protected returns remove directions, and only continuation constraints keep future utility and cost.

Four-action witness

	p	d	q = p+0.02d	Q _U	Q _C	s
a1	0.25	+1	0.27	1.0	1	+1
a2	0.25	-3	0.19	0.8	2	-1
a3	0.25	+3	0.31	0.6	2	+1
a4	0.25	-1	0.23	0.4	1	-1

utility 0.70 $\rightarrow$ 0.70 · cost 1.50 $\rightarrow$ 1.50 · key gain 0.16 · KL 0.0161 nats

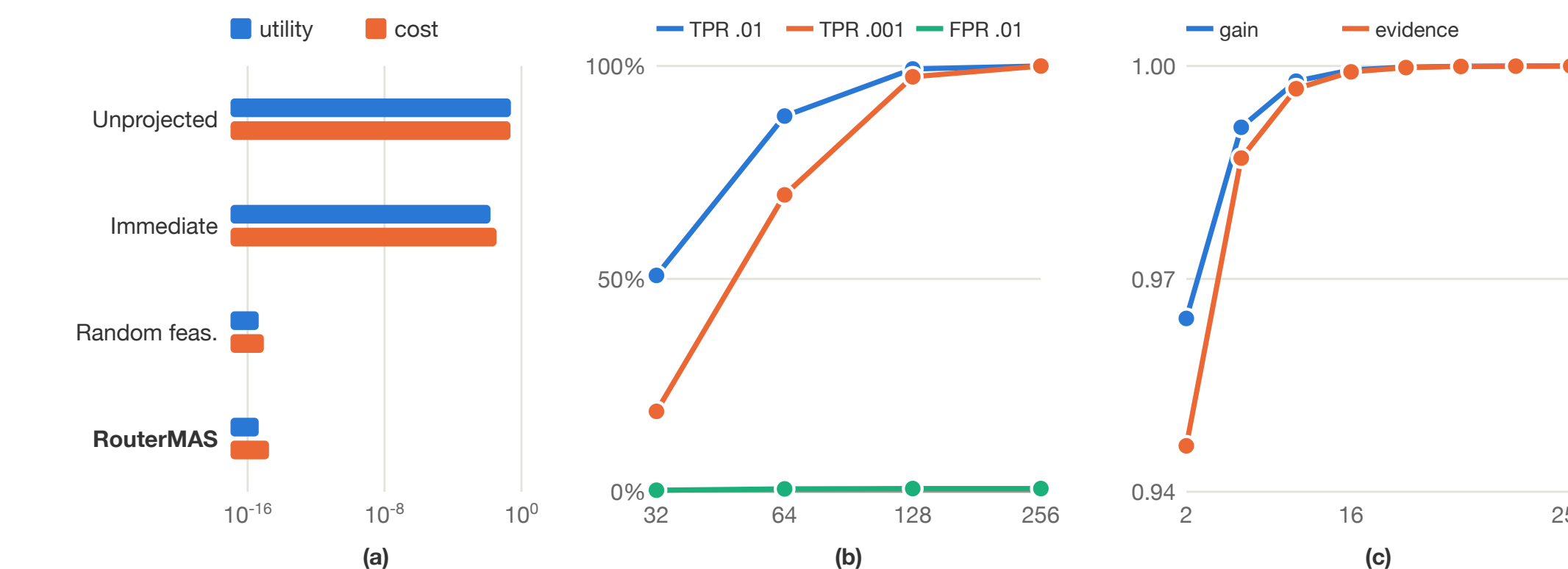
$d = (1, -3, 3, -1)$ satisfies $1^\top d = Q_U^\top d = Q_C^\top d = 0$: probability moves, utility and cost do not.

4 Controlled evidence, exact dynamics

Multi-agent routing system with known transitions: 4 delegation steps $\times$ 5 states, 6 legal delegations, 2 protected returns, exact dynamic programming, $\eta = 1.5$, seed 20260923. All numbers re-executed.

Worst prefix value drift 0.243 $\rightarrow$ <10⁻¹⁴ routing still moves: 0.093 nats / decision	Immediate-reward matching only 0.0156 / 0.0352 downstream utility / cost drift
Detection at $\alpha = 0.01$ 50.8% $\rightarrow$ 100% 32 $\rightarrow$ 256 decisions · FPR $\leq$ 0.75%	Embedding $\leftrightarrow$ evidence coefficient 0.946 $\rightarrow$ 0.99999 $\eta = 2 \rightarrow$ 256

Routing law	max $ \Delta V_U $	max $ \Delta V_C $	mean KL
Unprojected key bias	0.2432	0.2304	0.1666
Immediate-value	0.0156	0.0352	0.0898
Random feasible	<10 ⁻¹⁴	<10 ⁻¹⁴	0.0065
Row-space key	<10 ⁻¹⁴	<10 ⁻¹⁴	0
Full-rank (3 actions)	0	0	0
RouterMAS	<10⁻¹⁴	<10⁻¹⁴	0.0929



Evaluation protocol on LLM multi-agent systems

- **Tasks:** GSM8K · HumanEval · MBPP · ALFWorld; six role-specialized Qwen2.5-7B-Instruct delegates, Llama-3.1-8B-Instruct transfer; centralized, layered and graph-routed routers.
- **Comparisons:** unmarked, red/green delegation bias, behavioral-mark adaptation, forced inter-agent path, mechanism controls.
- **Metrics:** success / pass@1, normalized cost, prefix drift $D_{U,95}$, TPR at validated FPR for $T \in \{32, 64, 128, 256\}$.
- **Falsifiable:** nonzero held-out routing freedom (else abstain); detection within 1 pp success and 3% cost; Corollary 1 coefficients $\rightarrow$ 1 under estimated values.

Takeaways

- **Carrier = freedom left by the task values.** Bellman conservation turns a whole-future contract into local linear equations.
- **Abstention is correct.** Full-rank histories and row-space keys return the reference router exactly.
- **One projection, two roles.** The same component sets the keyed update and its verification information.